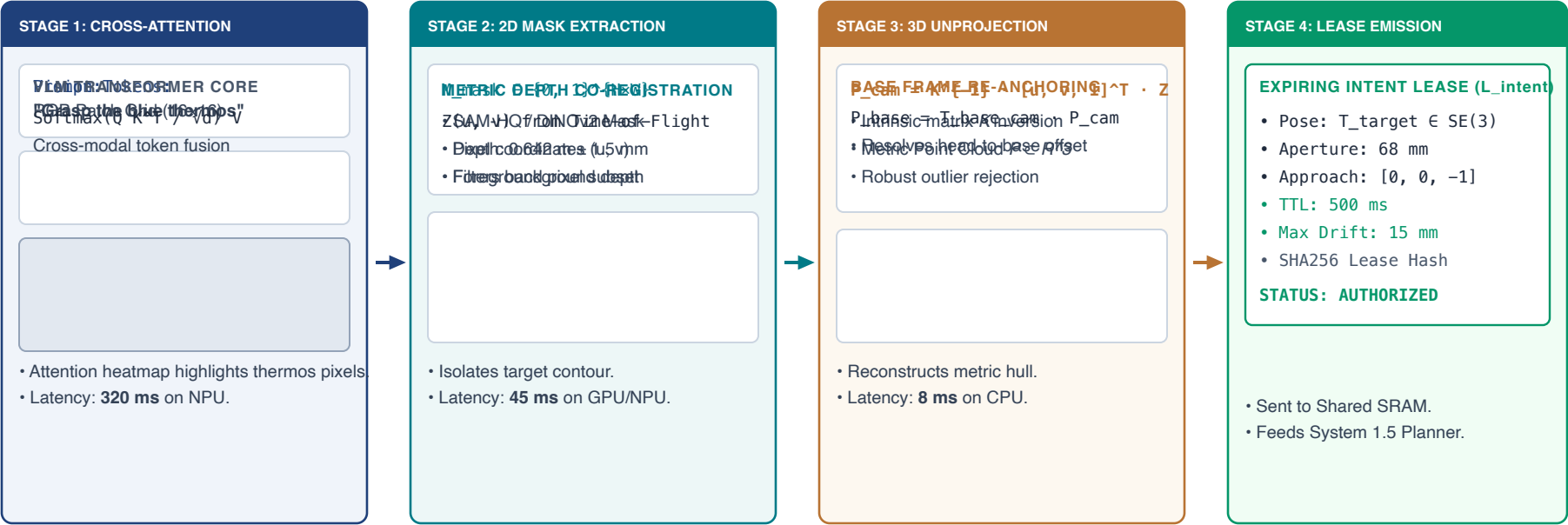


THE 4-STAGE VLM VISUAL & SPATIAL AFFORDANCE GROUNDING PIPELINE

Lifting open-vocabulary natural language prompts into metric SE(3) grasp targets and expiring intent leases



THE PHYSICAL AI INTENT LEASE CONTRACT (UNTRUSTED PROPOSAL → DETERMINISTIC VERIFICATION)

1. **Untrusted Origin:** The VLM runs on Linux MPU and can suffer GC stalls, memory spikes, or hallucinations. Its output is merely a *candidate proposal*.
2. **Expiring Lease Structure:** Every intent lease carries a strict time-to-live ($\tau_{TTL} = 500\text{ ms}$) and maximum scene drift threshold ($\delta_{drift} \leq 15\text{ mm}$).
3. **Microcontroller Gatekeeping:** The 1 kHz Reflex Brain on the MCU audits kinematic reachability and CBF invariants before a single motor microstep executes.